

Apache Maven & Java Build Automation

A Comprehensive Technical Reference Guide from Fundamentals to Advanced Lifecycle Management

Timestamp: 00:44

What is a JAR?

A **JAR (Java Archive)** is a package file format used to aggregate many Java class files, associated metadata, and resources into a single compressed file for distribution.

Timestamp: 04:59

Library vs. Application

Java Library (Dependency)

A collection of reusable behavioral classes design to be imported by other programs without an execution entry point.

Java Application (Executable)

A complete, functional program containing an execution entry point (main method).

Timestamp: 12:41

Introducing Maven

Apache Maven is a build automation tool based upon the concept of a Project Object Model (POM). It eliminates "Jar Hell" by automating dependency resolution and establishing unified structures.

Timestamp: 19:22 & 21:17

Maven Projects & Standard Folder Structure

Maven mandates a strict layout according to the concept of "*Convention over Configuration*".

```
my-maven-project/
├── pom.xml           <-- Configuration descriptor
└── src/
    ├── main/java/   <-- Production code execution logic
    └── test/java/    <-- Unit tests
```

Timestamp: 32:01 & 34:55

Compiling Code, the Target Folder, & Creating JARs

The build output lifecycle compiles files directly into a dynamically generated workspace named `target/`.

Timestamp: 36:30, 44:18 & 53:18

Understanding pom.xml & Dependencies

The `pom.xml` acts as the declarative manifest file configuring the dependencies and runtime target environments.

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POM Trees & Repository Management

Repository	Location Context	Description & Purpose
Local Repository	Stored inside your machine (<code>~/.m2/repository</code>).	Acts as a local cache layer to prevent repetitive network requests.
Central Repository	Hosted remotely by the Apache Community.	The global registry holding public open-source coordinates.

Timestamp: 01:11:47 & 01:24:29

The Maven Lifecycle & Archetypes

Phases systematically scale upward: **Validate** → **Compile** → **Test** → **Package** → **Verify** → **Install** → **Deploy**. Archetypes function as standard project templates to bootstrap uniform application configurations instantly.

untitled1 Version Control

Project

- untitled1 C:\Users\NITISH SHUKLA\IdeaProjects\untitled1
 - .idea
 - .mvn
 - src
 - main
 - java
 - org.example
 - Main
 - resources
 - test
 - target
 - .gitignore
 - pom.xml
 - External Libraries
 - temurin-25 C:\Program Files\Eclipse Adoptium\jdk-25.0.3
 - Maven: com.google.protobuf:protobuf-java:4.31.1
 - Maven: com.mysql:mysql-connector-j:9.5.0
 - Maven: com.sun.xml.bind:jaxb-impl:4.0.4
 - Maven: jakarta.activation:jakarta.activation-api:2.1.4
 - Maven: jakarta.inject:jakarta.inject-api:2.0.1
 - Maven: jakarta.persistence:jakarta.persistence-api:3.2.0
 - Maven: jakarta.transaction:jakarta.transaction-api:2.0.1
 - Maven: jakarta.xml.bind:jakarta.xml.bind-api:4.0.4

Main.java pom.xml (untitled1)

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  <dependencies>
    <dependency>
      <groupId>org.hibernate.orm</groupId>
      <artifactId>hibernate-core</artifactId>
      <version>7.5.7.Final</version>
      <scope>compile</scope>
    </dependency>
  </dependencies>
  <properties>
    <author.name>Nitish</author.name>
    <maven.compiler.source>25</maven.compiler.source>
    <maven.compiler.target>25</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
</project>
```

Maven

- package
- verify
- install
- site
- deploy
- Plugins
- Dependencies
 - com.mysql:mysql-connector-j:9.5.0
 - com.google.protobuf:protobuf-java:4.31.1
 - org.hibernate.orm:hibernate-core:7.5.7.Final
 - jakarta.persistence:jakarta.persistence-api:3.2.0
 - jakarta.transaction:jakarta.transaction-api:2.0.1
 - org.jboss.logging:jboss-logging:3.5.0.Final
 - org.hibernate.models:hibernate-models:7.5.7.Final
 - net.bytebuddy:byte-buddy:1.18.8
 - jakarta.xml.bind:jakarta.xml.bind-api:4.0.4
 - org.glassfish.jaxb:jaxb-runtime:4.0.5
 - jakarta.inject:jakarta.inject-api:2.0.1
 - org.antlr:antlr4-runtime:4.13.2
- Repositories

Run untitled1 [package]

```
[INFO] Building jar: C:\Users\NITISH SHUKLA\IdeaProjects\untitled1\target\untitled1-1.0-SNAPSHOT-jar.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.007 s
[INFO] Finished at: 2026-06-23T12:03:32+05:30
[INFO] -----
```

untitled1 pom.xml

38°C Sunny

22:1 LF UTF-8 4 spaces

12:13 23/06/2026

demo Version Control

Project

- .mvn
- src
 - main
 - test
 - java
 - com.example.demo
 - DemoApplicationTests
 - HelloControllerTest
- target
 - classes
 - generated-sources
 - generated-test-sources
 - maven-archiver
 - maven-status
 - surefire-reports
 - test-classes
 - demo-0.0.1-SNAPSHOT.jar
 - demo-0.0.1-SNAPSHOT.jar.original
- .gitattributes
- .gitignore
- HELP.md

pom.xml (demo) demo-effective-pom.xml spring-boot-starter-parent-4.0.7.pom

Rainbowify is disabled because the file is too large [got it, don't show again](#) [open setting](#)

```
13 <project xmlns="http://maven.apache.org/POM/4.0.0" >
38 <scm>
42 <url/>
43 </scm>
44
45 <issueManagement/>
46
47 <properties>
48 <activemq.version>6.1.8</activemq.version>
49 <angus-mail.version>2.0.5</angus-mail.version>
50 <artemis.version>2.43.0</artemis.version>
51 <aspectj.version>1.9.25.1</aspectj.version>
52 <assertj.version>3.27.7</assertj.version>
53 <awaitility.version>4.3.0</awaitility.version>
54 <brave.version>6.3.1</brave.version>
55 <build-helper-maven-plugin.version>3.6.1</build-helper-maven-plugin.version>
56 <
57 < Property
58 < build-helper-maven-plugin.version: 3.6.1
59 <
60 < cassandra-driver.version>4.19.3</cassandra-driver.version>
61 <classmate.version>1.7.3</classmate.version>
```

Maven

- Profiles
- demo
 - Lifecycle
 - clean
 - validate
 - compile
 - test
 - package
 - verify
 - install
 - site
 - deploy
 - Plugins
 - Dependencies
 - Repositories

Run DemoApplication x demo [package] x

demo [package]: At 23/06/2026 22:32 47 sec, 113 ms

```
[INFO] Repackaging main artifact D:\DEVELOPMENT\materials\BACKEND (JAVA)\JAVA backend\SpringBoot\demo-0.0.1-SNAPSHOT.jar from demo-0.0.1-SNAPSHOT.jar.original to demo-0.0.1-SNAPSHOT.jar
[INFO] The original artifact has been renamed to D:\DEVELOPMENT\materials\BACKEND (JAVA)\JAVA backend\SpringBoot\demo-0.0.1-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 44.829 s
[INFO] Finished at: 2026-06-23T22:32:09+05:30
[INFO] -----
```

demo demo pom.xml 13:40 CRLF UTF-8 4 spaces

Hot days ahead 30°C

Search

ENG IN 22:34 23/06/2026

Apache Maven is a build tool for Java projects. Using a project object model (POM), Maven manages a project's compilation, testing, and documentation.

This site is separated into the following sections:

Each guide is divided into a number of trails to get you started on a particular topic, and includes a reference area and a "cookbook" of common examples. You can access the guides at any time from the left navigation. If you are looking for a quick reference, you can use the [documentation index](#).

Support for Maven is available in a variety of different forms.